Capstone Report

Group 3

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Abstract

Multi-agent large language model systems often rely on one model to critique another, but a revised answer alone does not reveal how the receiving model internally processes that feedback. We study feedback uptake in a Solver–Critic system using MuSiQue, a multi-hop question-answering benchmark with known answers. A Gemma 3 4B instruction-tuned Solver first answers each question, a Qwen3 4B instruction-tuned Critic independently reviews that answer, and the Solver then chooses between its original response and the Critic's recommendation. We capture Solver residual-stream activations across nine transformer layers and use sparse autoencoders (SAEs) to represent those activations as sparse features. Linear probes identify features associated with adoption versus retention, while held-out causal interventions test whether altering selected features changes the Solver's decision. The collected candidate-presentation dataset contains 22,355 sampled questions; the current labeling policy identifies 5,834 adoptions, 4,810 retentions, 8,451 agreements, and 3,260 episodes excluded from the primary behavioral analysis. This design separates predictive association from causal evidence and provides a reproducible framework for investigating when language models defer to, resist, or agree with another model's feedback.

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1 Introduction

Large language models (LLMs) are increasingly deployed as components of multi-agent systems in which models critique, debate, verify, or revise one another's outputs. These systems can improve factuality and reasoning by combining multiple model responses [3]. Their success, however, depends on more than whether a Critic can produce useful feedback. A Solver must also decide whether that feedback should change its answer. Blind deference can propagate incorrect recommendations, while indiscriminate resistance can prevent useful correction. Prior work on self-correction and sycophancy shows that externally prompted revision does not necessarily imply reliable error correction [6, 8].

Behavioral evaluation can reveal whether a Solver changed its answer, but it cannot by itself identify the internal computations responsible for that decision. A representation may correlate with feedback adoption because it encodes disagreement, answer content, Critic quality, or a downstream consequence of the decision. Establishing a mechanism therefore requires two steps: finding internal features that predict behavior on held-out data and intervening on those features to test whether they influence behavior.

Sparse autoencoders (SAEs) offer a practical representation for this analysis. They decompose dense neural activations into a larger set of sparsely active features, some of which can be more interpretable than individual model dimensions [2, 4]. We use SAEs to encode the Solver's hidden states while it processes Critic feedback. Regularized linear probes then test whether the resulting feature activations predict whether the Solver adopts the Critic's recommendation or retains its original answer. Finally, selected features are ablated or modified during a replayed forward pass, allowing us to compare targeted interventions with random-feature controls.

Our experiments use MuSiQue, a benchmark designed to require composition over multiple reasoning steps [9]. For each question, a Gemma 3 4B instruction-tuned Solver produces an initial answer and a Qwen3 4B instruction-tuned Critic independently produces a review and recommendation. The Solver is subsequently shown its prior answer and the Critic's candidate answer in randomized order. This candidate-presentation protocol makes adoption and retention more directly observable than unconstrained free-form revision while preserving agreement and unresolved cases as separate outcomes.

This project addresses three primary research questions:

1. Which internal Solver representations predict adoption versus retention of Critic feedback?
2. Do interaction-trained SAEs isolate feedback-related structure more effectively than a general pretrained SAE?
3. Do targeted interventions on probe-selected SAE features causally change feedback-uptake behavior relative to appropriate controls?

The main contribution is an end-to-end, reproducible workflow connecting behavioral data collection, leakage-aware partitioning, sparse representation learning, predictive probing, and causal intervention. The workflow also retains agreement, refusal, ambiguous, and malformed responses rather than forcing every interaction into a binary target. This distinction is important: a model cannot meaningfully adopt a Critic's answer when both agents already agree, and uncertain text matches should not be treated as ground-truth behavior labels.

2 Related Work

2.1 Multi-agent critique and model revision

Multi-agent LLM methods use interaction among model instances to improve a final response. Multi-agent debate, for example, allows agents to exchange arguments and revise their answers over several rounds, producing gains on some factual and reasoning tasks [3]. A Solver–Critic system is a simpler instance of the same idea: one model proposes an answer and a second model evaluates it before revision. These approaches are motivated by the possibility that one agent can expose errors missed by another.

Revision is not automatically equivalent to correction. Huang et al. show that LLMs can struggle to improve reasoning through intrinsic self-correction without reliable external feedback [6]. Conversely, models may accept external signals for reasons unrelated to their correctness. Work on sycophancy documents cases in which models adapt responses to a user's stated view even when doing so reduces truthfulness [8]. Together, these findings motivate separating three questions: whether feedback is correct, whether the Solver adopts it, and whether adoption improves the final answer. Our primary behavioral target concerns adoption rather than correctness, while correctness is retained for stratified analysis.

2.2 Sparse autoencoders for language-model representations

Transformer residual streams are high-dimensional and superposed: a single dimension need not correspond to a single human-interpretable concept. SAEs learn an overcomplete dictionary that reconstructs dense activations from a sparse set of latent features. Cunningham et al. demonstrated that SAE features can be substantially more interpretable than raw neuron activations [2]. Subsequent work studied how SAE design and scale affect sparsity, reconstruction quality, and downstream interpretability [4].

SAE features remain learned model components rather than guaranteed semantic units. Reconstruction error, inactive features, correlated latents, and feature splitting can all complicate interpretation. For that reason, this study compares multiple SAE variants and measures both reconstruction behavior and downstream predictive utility. A general pretrained SAE tests whether broadly learned features transfer to feedback processing, whereas interaction-trained SAEs test whether training on the experiment's activation distribution better captures the relevant structure.

2.3 Probing and causal validation

Linear probes test whether information can be decoded from an intermediate representation [1]. Their simplicity makes coefficients useful for ranking candidate features, but probe performance alone does not show that the model uses those features to produce its behavior. Probes may exploit correlated information or properties of the dataset rather than a causal mechanism. Control tasks, held-out evaluation, and careful capacity selection are therefore necessary when interpreting probe results [5].

Feature intervention provides a stronger test. Work on sparse feature circuits demonstrates how sparse features can support both circuit discovery and targeted model editing [7]. Following this general principle, our analysis uses probe coefficients only to nominate candidate SAE features. We then replay held-out Solver interactions while ablating selected features and compare the resulting adoption rate with no-intervention and random-feature controls. This progression from decoding to intervention is intended to distinguish features that merely describe feedback uptake from features that help produce it.

2.4 Multi-hop question answering as an experimental setting

MuSiQue constructs multi-hop questions by composing connected single-hop questions and filtering shortcuts that allow models to bypass the intended reasoning chain [9]. It is useful for the present study because it supplies question identifiers, supporting passages, known answers, and answer aliases at sufficient scale for repeated Solver–Critic interactions. Its range of two- to four-hop questions also creates variation in difficulty. At the same time, exact or normalized string matching cannot resolve every semantic equivalence. Our labeling procedure therefore uses question-local aliases and conservative text rules, and excludes cases whose adoption status cannot be determined reliably.

3 Exploratory Data Analysis

3.1 Dataset construction

The current candidate-presentation collection contains 22,355 sampled MuSiQue questions. Each episode follows the same sequence: the Solver produces an initial answer (A1), the Critic independently reviews that answer and supplies a recommendation (C1), and the Solver produces a second answer (A2) after both candidates are presented in randomized order. Generation is deterministic for the recorded run. Solver activations are captured at decoder layers 4, 8, 17, 25, 29, 30, 31, 32, and 33 so that later analyses can compare where feedback-related behavior is most decodable.

Questions are assigned to frozen train, validation, test, and intervention partitions by question identifier. All records associated with one question remain in the same partition. The training partition is used to fit SAEs and probes, validation supports model and feature selection, test is reserved for final predictive evaluation, and the intervention partition is reserved for causal experiments. This question-level separation prevents related interactions from appearing on both sides of an evaluation boundary.

3.2 Behavior-label distribution

The labeling policy compares A1, C1, and A2 using normalized text, question-specific answer aliases, and conservative rules for refusals and ambiguous responses. Each episode receives one of four mutually exclusive labels:

- **Adoption**: A1 and C1 differ, and A2 follows the Critic.
- **Retention**: A1 and C1 differ, and A2 retains A1.
- **Agreement**: A1 and C1 agree, and A2 preserves that answer.
- **Out**: the behavior cannot be compared reliably or the episode fails an inclusion rule.

Adoption and retention together form the primary binary cohort of 10,644 episodes, or 47.6% of the full collection. Within this cohort, adoption accounts for 54.8% and retention for 45.2%. This is substantially more balanced than an earlier free-form-feedback labeling approach, which was dominated by adoption. Nevertheless, class-aware metrics such as balanced accuracy, per-class precision and recall, F1 score, and area under the ROC curve remain preferable to raw accuracy.

The label proportions are broadly similar across the four frozen partitions, which reduces the risk that an observed evaluation difference is driven only by a major shift in the outcome distribution. The 23 unassigned questions are quarantined as *Out* and are not used for model fitting or evaluation.

Table 1: Behavior labels in the candidate-presentation collection.

Label	Episodes	Share
Adoption	5,834	26.1%
Retention	4,810	21.5%
Agreement	8,451	37.8%
Out	3,260	14.6%
Total	22,355	100.0%

Table 2: Behavior labels by frozen experimental partition.

Partition	Adoption	Retention	Agreement	Out
Train	4,146	3,465	6,063	2,262
Validation	625	491	901	396
Test	529	416	755	292
Intervention	534	438	732	287
Unassigned	0	0	0	23

3.3 Data-quality observations

Several response patterns require special handling. The collection includes 1,060 episodes in which both A1 and the Critic refuse or reject the premise; these are generally agreements rather than adoption events. Other ambiguous patterns include letter-only candidate selections, citation-only replies, hyphenation differences, hybrid answers that combine both candidates, and Critic responses that repeat A1 while nominally disagreeing. The final rules retain explicit flags for these cases and exclude uncertain comparisons from the primary binary target.

These observations constrain the interpretation of downstream probes. A probe could learn answer length, refusal style, candidate position, or Critic correctness instead of an internal tendency to accept feedback. Downstream analysis should therefore include text-only and majority-class baselines, report results by relevant behavioral flags, and verify that candidate order is not predictive of the label. Probe-selected features should be interpreted as correlates until targeted intervention demonstrates an effect beyond random-feature and reconstruction controls.

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